Day 02: Variables and Data Types

1. Introduction

Variables are used to store data in Python. Every variable has a name and stores a value that can be used later in the program.

Example:

name = "raju"

age = 20

print(name)

print(age)

2. Rules for Naming Variables

Variable names should start with a letter or underscore (_).

Variable names cannot start with a number.

Variable names are case-sensitive.

Spaces are not allowed.

Use meaningful variable names.

Examples:

student_name = "Rahul"

marks = 95

Invalid Examples:

2name = "Rahul"

student name = "Rahul"

3. Data Types in Python

Python supports different data types.

Integer (int)

Stores whole numbers.

Example:

age = 20

Float (float)

Stores decimal numbers.

Example:

height = 5.7

String (str)

Stores text.

Example:

name = "Python"

Boolean (bool)

Stores True or False values.

Example:

is_student = True

4. Type Checking

The type() function is used to check the data type.

Example:

x = 100

print(type(x))

Output:

<class 'int'>

5. Type Conversion

Python allows converting one data type into another.

Example:

age = "20"

print(int(age))

6. Taking Input from User

The input() function accepts data from the user.

Example:

```
name = input("Enter your name: ")


print(name)
```

7. Multiple Variables

Example:

```
a, b, c = 10, 20, 30


print(a)

print(b)

print(c)
```

8. Practice Programs

Program 1

```
name = "alexa"


print(name)
```

Program 2

```
age = 20


print(type(age))
```

Program 3

```
marks = float(input("Enter marks: "))


print(marks)
```

Key Points

Variables store data.

Python has int, float, string and boolean data types.

Use type() to check the data type.

Use input() to receive user input.

Use type conversion when required.

Summary

In Day 02, we learned:

Variables

Naming rules

Integer

Float

String

Boolean

Type checking

Type conversion

User input